

Scenario-Based Interview Questions – Constraint

1. Employee Registration System (PRIMARY KEY & AUTO_INCREMENT)

Scenario:

You are designing an Employee Management System for a company. Every new employee must have a unique Employee ID, and the application should automatically generate this ID without manual entry.

Interview Question:

Which SQL constraint(s) would you use to ensure each employee gets a unique, automatically generated Employee ID, and why?

Answer : primary Key and auto-increment because, unique constraint used to remove or avoid duplicates and Auto-increment helps to generate employee ID automatically

Query : `Create table Employee(emp Id int auto increment primary Key);`

2. Online Banking Application (UNIQUE Constraint)

Scenario:

Abank allows customers to register for internet banking. Each customer must register with a unique email address, and duplicate email IDs should not be allowed.

Interview Question:

Which SQL constraint would you use to prevent duplicate email addresses in the Customer table? Explain your choice.

Answer : Unique, because unique used to avoid duplicates

Query : `create table Custome`

`r ( Customer Id int, name varchar(15), email varchar (20) unique);`

3. College Management System (FOREIGN KEY Constraint)

Scenario:

Acollegehas two tables: **Department** and **Student**. Every student must belong to an existing department. Studentsshould not be assigned to a department that does not exist.

Interview Question:

Which SQL constraint would you use to maintain the relationship between the Student and Department tables and prevent invalid department assignments?

Answer : Foreign key used to create a relationship between a tables

Query : `create table student (foreign key (stuID) reference department(departmentID);`

4. E-Commerce Order Management (CHECK Constraint)

Scenario:

Ane-commerce company stores product prices in the database. The business rule states that the product price should always be greater than zero. Negative or zero prices should never be accepted.

Interview Question:

Which SQL constraint would you use to enforce this business rule and prevent invalid product prices?

Answer : Check used to validated the conditions

Query : create table product (productID int, product name varchar(15),
product prize varchar (30) check(product prize > 0));

5. Hospital Patient Registration (NOT NULL & DEFAULT Constraint)

Scenario:

A hospital is developing a Patient Registration System. Every patient must have a name, but if the patient's blood group is not entered during registration, it should automatically be stored as 'Unknown'

Interview Question:

Which SQL constraints would you use to ensure the patient name is always provided and a default blood group is assigned when no value is entered?

Answers : Not null used to avoid nulls

Default used to assign a default value

Create patient registration(
patient ID int,
patient name varchar (20) not null,
bloodgroup default (unknown));